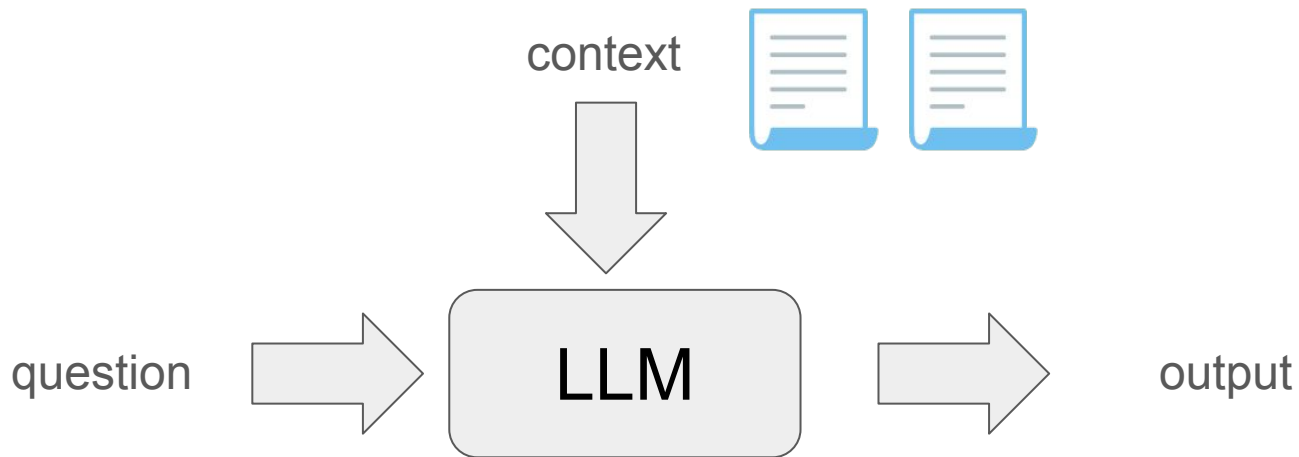


Noise-Gated Retrieval for Cost-Aware RAG

Arnold Alvaro Moya Lavado

RAG



Purpose: Reduce hallucination

HotpotQA

What government position was held by the woman who portrayed Corliss Archer in the film Kiss and Tell?

```
{  
  "title": [  
    "Meet Corliss Archer",  
    "Shirley Temple",  
    "Janet Waldo",  
    "Meet Corliss Archer (TV series)",  
    "Lord High Treasurer",  
    "A Kiss for Corliss",  
    "Kiss and Tell (1945 film)",  
    "Secretary of State for Constitutional Affairs",  
    "Village accountant",  
    "Charles Craft"  
  ],  
}
```

HotpotQA

```
],  
"sentences": [  
  [  
    "Meet Corliss Archer, a program from radio's Golden Age, ran from January 7, 1943 to September 30, 1951.",  
    " Although it was CBS's answer to NBC's popular \"A Date with Judy\", it was also broadcast by NBC in 1949.",  
    " From October 3, 1952 to June 26, 1953, it aired on ABC, finally returning to CBS.",  
    " Despite the program's long run, fewer than 24 episodes are known to exist."  
  ],  
  [  
    "Shirley Temple Black (April 23, 1928 - February 10, 2014) was an American actress, singer, dancer, and comedian.",  
    " As an adult, she was named United States ambassador to Ghana and to Czechoslovakia and also served as a UNICEF ambassador.",  
  ],  
  [  
    "Janet Marie Waldo (February 4, 1920 - June 12, 2016) was an American radio and voice actress.",  
    " She is best known in animation for voicing Judy Jetson, Nancy in \"Shazzan\", Penelope Pitstop, and the character of the Queen of the Jungle in \"The Jungle Bunch\".",  
  ],  
  [  
    "Meet Corliss Archer is an American television sitcom that aired on CBS (July 13, 1951 - August 10, 1953).",  
    " The program was an adaptation of the radio series of the same name, which was based on a series of short stories by Charles G. Smith.",  
  ],  
  [  
    "The post of Lord High Treasurer or Lord Treasurer was an English government position and has been a part of the Royal Household since the 14th century.",  
    " A holder of the post would be the third-highest-ranked Great Officer of State, below the Lord High Chamberlain and the Lord High Steward."
```

Research question

Can lightweight re-ranking systems achieve token savings in fixed-pool multi-hop QA without degrading reasoning quality, and what benchmark properties determine success or failure?

Motivation - The cost problem

- Most RAG systems pass a fixed top-k documents to the LLM on every query regardless of how many are actually useful <https://arxiv.org/abs/2005.11401>
- On HotpotQA distractor with GPT-5.4-mini, fixed k=10 costs 1,452 tokens/query — roughly 14× more than answering without retrieval (103.5 tokens)
- Noisy or irrelevant context actively degrades answer quality, not just wastes tokens <https://aclanthology.org/volumes/2026.eacl-industry/>
- At scale, this per-query overhead becomes a significant inference-cost bottleneck

Why Context Selection Is Hard on Multi-Hop QA

- Distractors are not random noise — they are BM25-retrieved precisely because they are topically adjacent to gold evidence, making them semantically hard to distinguish <https://aclanthology.org/D18-1259/>
- Multi-hop reasoning requires jointly complementary evidence sets, not just individually relevant passages
<https://aclanthology.org/volumes/2023.acl-long/>
- The fixed-pool structure (2 gold + 8 distractors) creates a structured filtering challenge, not just a ranking problem

Research GAP

- Training-based methods (Self-RAG, DioR, DRAGIN) require model modification or additional supervision: impractical for test-time deployment
- Chunk-level methods (ChunkRAG) require extra infrastructure for boundary detection: high integration overhead
- Gap-based truncation (Adaptive-k) assumes similarity rank is monotone in relevance: this assumption breaks on distractor-heavy benchmark
- Learned bandit strategies (UCB1-TUNED, LinUCB) are limited by sparse title vocabulary and lexically confusable features — both fail structurally on HotpotQA

Goals

- Propose a lightweight re-ranking, training-free filter that preserves top-k=10 answer quality while reducing token cost
- Explain why some approaches work better than others through structural failure mode analysis of learned strategies on distractor-heavy benchmarks

Experiments

Fixed baselines ($k=0, 3, 5, 10$): establish the quality-cost spectrum from parametric-only to full retrieval, ranging from $F1=0.387$ to $F1=0.792$; these anchor all comparisons in Table 2.

Type	Strategy	F1	EM	Tokens	Savings	Tolerance
Baselines	No retrieval ($k=0$)	0.387	0.278	103.5	-92.9%	×
	Fixed $k=5$	0.716	0.555	731.7	-49.6%	×
	Fixed $k=10$	0.792	0.632	1452.5	N/A	ref.



Reference

Even top models with full retrieval are not 100% performant

Experiments

Adaptive-k and learned strategies (UCB1-TUNED, LinUCB): evaluated against the fixed baselines in Table 2, all three fall short of even fixed $k=5$

Type	Strategy	F1	EM	Tokens	Savings	Tolerance
Baselines	No retrieval ($k=0$)	0.387	0.278	103.5	-92.9%	×
	Fixed $k=5$	0.716	0.555	731.7	-49.6%	×
	Fixed $k=10$	0.792	0.632	1452.5	N/A	ref
Learned	Adaptive- k	0.627	0.475	483.1	-66.8%	×
	UCB1-TUNED ($k=5$)	0.704	0.570	723.0	-50.2%	×
	LinUCB ($k=5$)	0.620	0.489	686.0	-52.8%	×

UCB-based methods try to find equilibrium between exploration and exploitation

Experiments

As Table 2 shows, **Noise-Gate is the only strategy** within the ± 0.05 F1 tolerance of $k=10$ (F1=0.764, -26.7% tokens), while all learned strategies fall well below: Adaptive- k (0.627), LinUCB (0.620), and UCB1-TUNED (0.704) despite comparable token budgets.

Type	Strategy	F1	EM	Tokens	Savings	Tolerance
Baselines	No retrieval ($k=0$)	0.387	0.278	103.5	-92.9%	×
	Fixed $k=5$	0.716	0.555	731.7	-49.6%	×
	Fixed $k=10$	0.792	0.632	1452.5	N/A	ref.
Learned	Adaptive- k	0.627	0.475	483.1	-66.8%	×
	UCB1-TUNED ($k=5$)	0.704	0.570	723.0	-50.2%	×
	LinUCB ($k=5$)	0.620	0.489	686.0	-52.8%	×
Noise-Gate	$\tau=0.2$	0.779	0.630	1317.5	-9.3%	✓
	$\tau=0.3$	0.764	0.603	1065.3	-26.7%	✓

Experiments - ablation

Noise-Gate τ sweep. Table 4 traces the full quality-cost frontier across five thresholds (0.2–0.5), identifying $\tau=0.3$ as the Pareto knee at 26.7% token reduction with F1 within tolerance of $k=10$.

τ	F1	EM	Tokens	Savings	Within tolerance?
0.20	0.779	0.630	1317.5	−9.3%	✓
0.25	0.765	0.603	1207.3	−16.9%	✓
0.30	0.764	0.603	1065.3	−26.7%	✓
0.35	0.726	0.570	889.8	−38.7%	×
0.50	0.619	0.490	403.9	−72.2%	×

Experiments - ablation

Jaccard redundancy ablation: Table 5 shows τ as the primary quality-cost lever, but ρ contributes meaningfully by removing near-duplicate paragraphs that pass the cosine gate, a necessary safeguard when topically adjacent passages cluster together in multi-hop settings. **Necessary for distractors**

Jaccard ρ	F1			Total Tokens (mean)		
	$\tau=0.2$	$\tau=0.3$	$\tau=0.5$	$\tau=0.2$	$\tau=0.3$	$\tau=0.5$
0.50	0.771	0.761	0.613	1307.5	1057.4	402.2
0.55	0.777	0.749	0.613	1308.3	1057.3	402.3
0.65	0.779	0.764	0.619	1317.5	1065.3	403.9

Experiments

Transfer pilots on 2WikiMultiHopQA and MuSiQue-Ans. Table 3 reports that Noise-Gate generalizes with **simultaneous quality and cost improvements** on both datasets, requiring at most one step of threshold recalibration and no retraining.

Dataset	Strategy	F1	EM	Tokens	Δ F1	Savings
2WikiMultiHopQA	Fixed $k=5$	0.526	0.460	638.4	-0.091	-38.6%
	Fixed $k=10$ (ref)	0.617	0.540	1040.1	—	—
	Noise-Gate $\tau=0.25$ ★	0.662	0.590	810.9	+0.045	-22.1%
	Noise-Gate $\tau=0.3$	0.605	0.550	708.0	-0.012	-31.9%
MuSiQue-Ans	Fixed $k=5$	0.469	0.360	741.6	-0.030	-44.7%
	Fixed $k=10$ (ref)	0.499	0.400	1342.0	—	—
	Noise-Gate $\tau=0.3$ ★	0.551	0.440	1193.2	+0.052	-11.2%
	Noise-Gate $\tau=0.2$	0.595	0.500	1826.3	+0.096	+36.0%

Slightly better F1 + token saving!

Failure modes

- **Adaptive-k**: the biggest similarity gap falls inside the **distractor** cluster, not at the evidence boundary, so it cuts in the wrong place.
- **UCB1-TUNED**: title vocabulary **barely overlaps between train and validation**, triggering cold-start on 83.8% of queries at $k=5$
- **LinUCB**: **distractors are BM25-retrieved** so they are lexically identical to gold paragraphs by construction, making the features useless

Very new data in validation and several distractors.
UCB requires fixed options to select. Like learn how
much explore based on the title or features.